

User Manual for Installation and Building of Project

Project Structure Overview

```
iot-chatbot/
├── backend/           # Flask API + Crawler
│   ├── app.py        # Main backend server
│   ├── embedding.py   # Embedding logic (MiniLM-L12-v2)
│   ├── chroma_setup.py # ChromaDB DB logic
│   ├── .env          # API keys, DB URIs, etc.
│   └── crawler/       # Scrapy Spider with BeautifulSoup
│       ├── iotscraper/
│       ├── scrapy.cfg
│       └── logs/
├── data/              # Scraped content files
├── chroma_db/         # Vector storage directory
├── frontend/          # Frontend (HTML/CSS/JS)
│   └── index.html
├── requirements.txt    # All Python dependencies
├── Dockerfile          # Backend image
└── docker-compose.yml  # Full deployment config
```

Prerequisites

Install the following on your system before running:

- Python 3.11+
- Node.js
- Docker and Docker Compose (for containerized setup)
- pip – Python package manager

Installation (Non-Docker)

Step-by-step for running locally:

Step 1: Clone the project

```
git clone https://github.com/Har-shan/iot-chatbot.git
cd iot-chatbot
```

Step 2: Create virtual environment

```
python -m venv venv
source venv/bin/activate    { On Windows: venv\Scripts\activate }
```

Step 3: Install dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
```

Step 4: Create .env file in backend/

Paste the following:

- TOGETHER_API_KEY=<your_together_api_key>
- MONGO_URI=<your_mongodb_atlas_uri>

(In repo its already there if you want can add your own api keys)

Step 5: (Optional) Crawling Data

```
python backend/crawler
Scrapy crawl iot_spider
```

(Since Git repo doesnt have chromadb You have to run embed_data.py)

Step 6: (Optional) Embed data

```
python backend/embedding.py
```

Step 7: Start Flask backend

```
cd backend
python app.py
```

Step 8: Open frontend

```
cd ../frontend
open index.html    Or  open manually in browser
```

Installation via Docker:-

Build and start:

From root of project:

`docker compose up --build`

(While staring this make sure your running docker desktop)

To stop:

`CTRL+C`

`docker compose down`

It runs on:

localhost:5000 → Flask backend

Crawling New Data (Manual Trigger)

Enter container shell

`docker exec -it iot-chatbot-backend bash`

Run scraper inside container

`cd crawler`

`scrapy crawl iot_spider`

->New data will be saved in /data/

(Since Git repo doesnt have chromadb You have to run `embed_data.py`)

Re-Embed New Data (After Crawling)

From project root (activate venv if needed)

`python backend/embedding.py`

This updates ChromaDB with new vectors.

And then start to build the docker.

Chatbot Usage

Ask questions through the web interface.

Chatbot retrieves relevant info using RAG (Retrieval-Augmented Generation).

Backend generates answer using Together AI LLM.

Feedback Logging

Each answer includes 👍👎 buttons:

Logs feedback into MongoDB Atlas (feedback collection)

Helpful for future analysis.

API Endpoints

Endpoint	Method	Description
/chat	POST	Sends user question, gets reply
/feedback	POST	Logs feedback for a response

Troubleshooting Tips

Issue		Fix
Empty context	=	Make sure data is scraped and embedding.py is run
API error / timeout	=	Check internet / Together API key
MongoDB SSL handshake error	=	Whitelist IP in MongoDB Atlas / fix .env URI
Docker not starting	=	Try docker compose down && docker compose up --build
Feedback not storing	=	Confirm MongoDB connected & collection created automatically

While running this project make sure docker desktop and mongodb atlas is running.

Details:-

Name: Rathod Harshan

E-mail: harshan.r21@iiits.in

Phone No: 9490420371

